

Then we have the outlet assembly which sits atop the copper plate and the shroud. As the name spells it out, the outlet assembly is where the coolant which has just made contact with the copper plate is drawn in and let out towards the radiator.

On top of the outlet assembly is the inlet assembly. This is where cool coolant is pumped onto the copper plate. The inlet assembly is where you'll see the impeller of the motor. The impeller is that part which provides suction so that the coolant can be drawn in. Coolant that is drawn in from the radiator is then forced into the cooling chamber.

The impeller interacts with the coolant directly and it is connected to the motor driver. Since the motor driver is electrically driven, you can't have water (or coolant) coming in contact with it. So in order to isolate the impeller from the motor driver and all electronic components, you have a housing. The housing has to be made of durable materials because even the motor on its own generates a bit of heat. So the coolant has to cool the CPU as well as the motor. This is why the standard instruction for any pump on the planet is that you should not run it while it is dry. The heat generated while doing so can burn the coils and render the motor useless.

On top of the motor driver is the housing which keeps the motor driver in place and also seals off the CPU block from the top. This is how the standard AIO functions. However, with RGB being the new trend, you need a little bit of razzle-dazzle. The RGB components are generally affixed on top of the motor housing with a bit of transparent casing or diffuser (or both) on top. The top housing is also where all the LED and pump wires are connected to.

Radiator

Radiators in AIOs aren't special in any way. They function in the same manner as any other radiator. Radiators consist of snaking tubes from one end to the other. These snaking tubes have a lot of fins attached to them which draws heat away from the tube and dissipates it into the air. In order to increase the surface area of the radiator, the fins have a wavy design and are densely packed together. The cooling fans are attached directly onto the radiator and are of high static pressure which ensures that more air pushes past the fins.

The metal used in the radiator shouldn't form a galvanic cell with the metal from the remaining components within the AIO. If it does, then whichever metal ends up being the anode will undergo corrosion faster than what's natural.